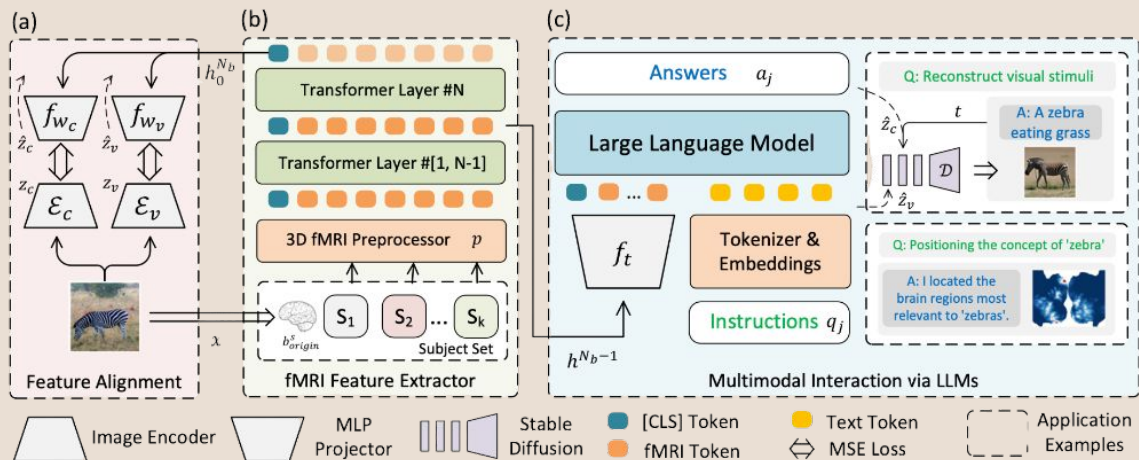
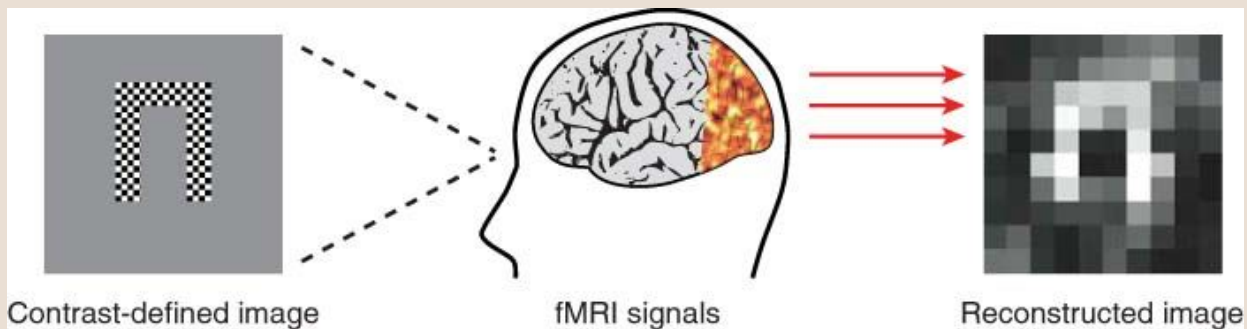


# **Neuro-Vision to Language:** Enhancing Brain Recording-based Visual Reconstruction and Language Interaction

**Authors:** Guobin Shen, Dongcheng Zhao, Xiang He, Linghao Feng, Yiting Dong, Jihang Wang, Qian Zhang, Yi Zeng

**Presented By:** Armina Fani

# Overview

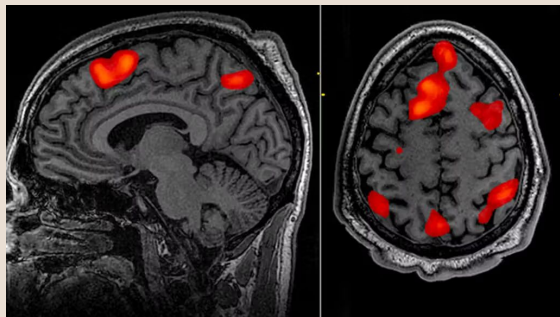


## Overview of the Method

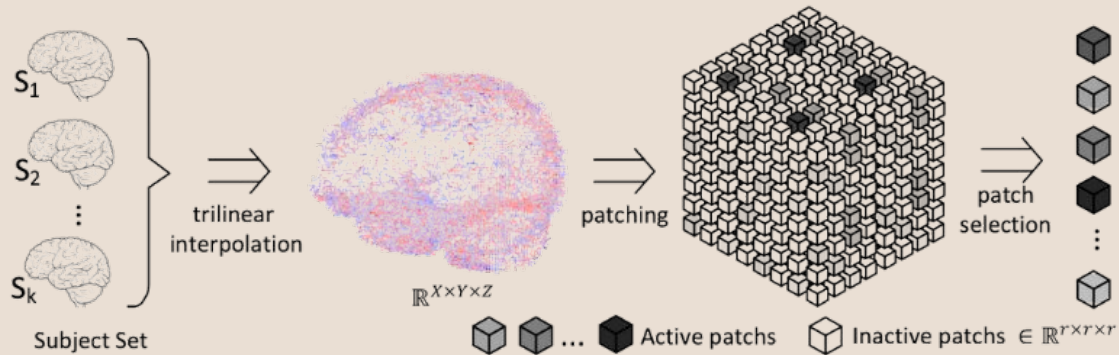
# fMRI Preprocessing

$$\{p : b_{\text{origin}}^s \mapsto b \mid b_{\text{origin}}^s \in \mathbb{R}^{X_s \times Y_s \times Z_s}, b \in \mathbb{R}^{N \times C}\}.$$

- **3D volume**: average BOLD response across 3 trials OR single-trial
- **Trilinear interpolation** → resize data to 83 x 104 x 81
- **Patching** → divide into 14 x 14 x 14 cubes
  - Remove patches with non-task-relevant information
- Flatten into a matrix of  $\mathbb{R}^{N \times C}$



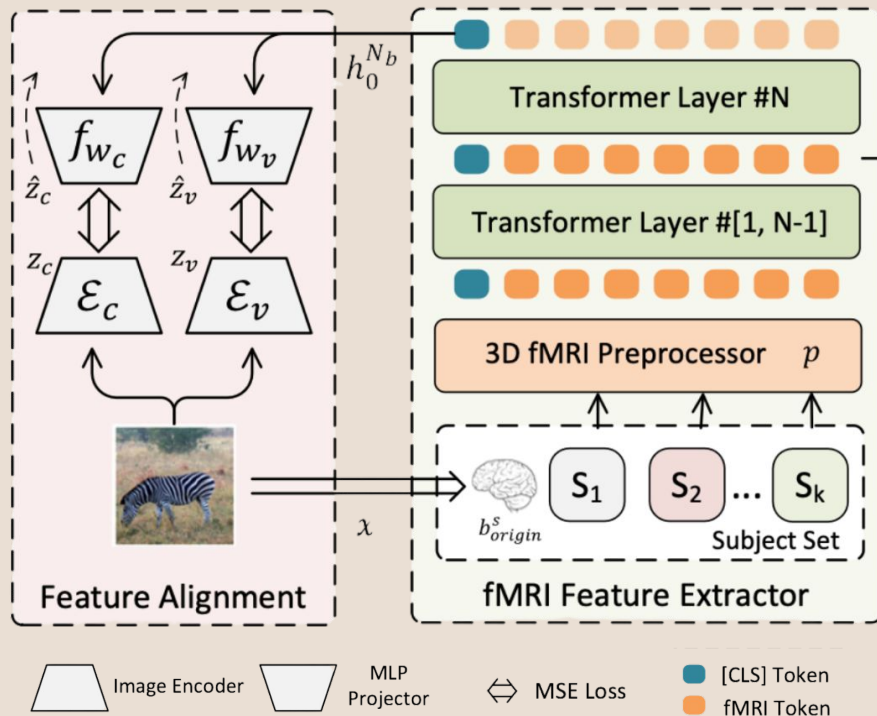
Example of BOLD fMRI



Overview of fMRI Preprocessing Workflow

# Dual-Stream fMRI Feature Extractor

- fMRI tokens are processed through a 16-layer Transformer Encoder (ViT3D-style) to extract features
  - **Hidden size:** 768
  - **Output:** [CLS] token from the last layer  $h_0^{N_b}$
- Two MLP heads align the [CLS] tokens with the stimulus image's:
  - **CLIP embeddings** ("this is a zebra")  $\mathcal{E}_c$
  - **VAE features** (stripes, layout of grass)  $\mathcal{E}_v$
- Feature extractor trained with MSE loss



Overview of Dual-Stream Feature Extractor

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(b,x) \sim P(B,X)} \left[ \left\| f_{w_c}(h_0^{N_b}) - \mathcal{E}_c(x) \right\|_2^2 + \alpha \left\| f_{w_v}(h_0^{N_b}) - \mathcal{E}_v(x) \right\|_2^2 \right].$$

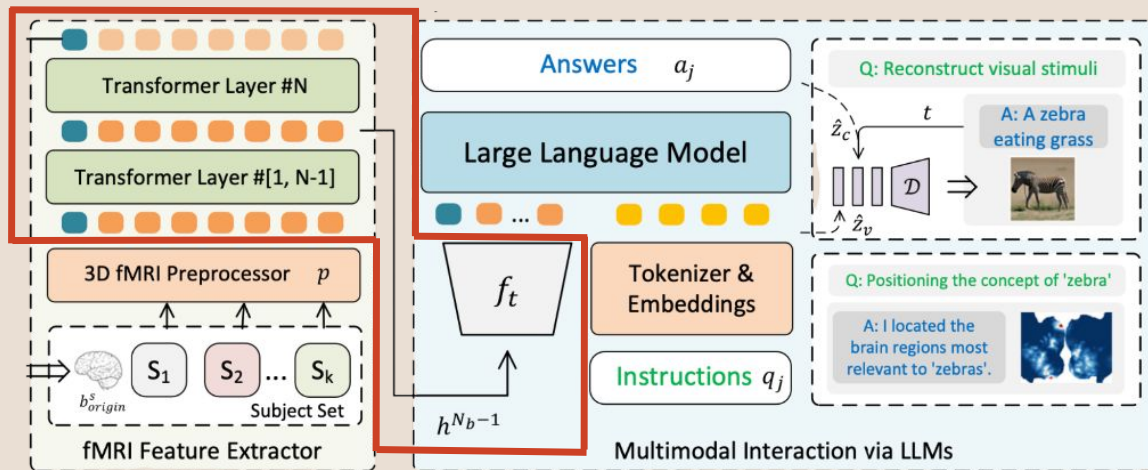
Feature Extractor Loss Function

# Multimodal Interaction with fMRI

**Goal:** Teach an LLM how to read fMRI signals as if they are words

**Problem:** The LLM (Llama-3-8B) cannot directly interpret raw vectors from the Transformer.

**Solution:** A two-layer perceptron maps the Hidden states from the second-to-last layer of the transformer to the same dimensions as LLM text embeddings.  $\bar{t} = f_t(h^{N_b-1})$



Overview of fMRI Projection

# Multimodal Interaction with fMRI

This mapping of fMRI tokens to the LLM is accomplished through a ***two-stage training strategy***:

## Stage 1:

- Freeze LLM
- Only tune fMRI token projector  $f_t$

## Stage 2:

- Fine-tune LLM and fMRI token projector  $f_t$  simultaneously

This training is done with the goal of maximizing the following objective function:

$$\max p_{\theta}(\mathbf{A}|\mathbf{Q}, t) = \max \prod_{j=0}^L p_{\theta}(a_j | q_j, a_{j-1}, \dots, q_0, t).$$

Objective Function: Maximize the probability of generating a sequence of answers  $\mathbf{A}$  given a sequence of questions  $\mathbf{Q}$  and the derived fMRI embeddings  $t$

# Interaction and Reconstruction via LLMs

Now that we have an LLM that understands fMRI signals + we have predicted CLIP embeddings and VAE latents, we can do several things:

Natural Language  
Interaction

Visual  
Reconstruction

Concept  
Localization

**Sample dialogue format:**

<human>: [fMRI data embedding] [instruction]

<bot>: [answer]

## Multi-round Dialogue

Q: <image>\n How many cows are grazing in the field?

A: There are about twenty cows Grazing.

Q: Can you describe the positioning of the cows?

For Reference



A: The cows are spread out in the field, with some cows closer to the trees while others are further away.

## Complex Reasoning

Q: <image>\n What is the person likely doing with the remote control based on their posture and the surrounding environment?

For  
Reference



A: The person is likely watching TV or controlling some electronic device with the remote control. The way they are holding it and the cozy setting with a blanket suggests a relaxed and leisurely activity.

## Visual Reconstruction

Q: <image>\n Provide the corresponding Stable Diffusion prompts for the image.

A: A man and a boy stand together in a field, flying a kite with a bright, colorful tail, amidst lush green grass and a clear blue sky.

For Reference

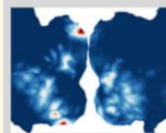


## Concept Locating

Q: <image>\n Locating the concept of "giraffe"

A: Locating the concept...

For  
Reference



# Experiments- Datasets

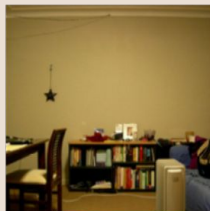
- Natural Scenes Dataset (NSD)
  - 7T fMRI scans and visual stimuli from COCO
  - 4 subjects
- Stimuli from COCO were cropped for fMRI presentation, which required re-annotation of the dataset.
  - New captions generated using BLIP2, and new bounding boxes with DETR



A person **kite** boarding in rough seas near the shoreline.



A jeep with a **dead bird** on the windshield.



There is a **laptop computer** in a room with a couch and a bookcase.



**three** people riding horses on a beach

Examples of annotation mismatch for cropped images

# Natural Language Interaction

- **Brain Captioning:** Generate a single concise sentence that summarizes the image
- **Detailed Description:** Provide an elaborate explanation of the scene
- **Complex Reasoning:** Infer something not explicitly present in the image



## Detailed Description

Q: `<image>`\n Describe the following image in detail.

A: a gray cat is peacefully resting under a white blanket on top of a bed. The cat's head is peeking out from underneath the blanket, illuminated by a gentle light. The cat seems comfortable and relaxed in its hiding spot. The bed occupies the lower portion of the image, while the cat takes up a significant portion of the upper area, creating a contrast between the soft bedding and the furry feline. The overall atmosphere is serene and domestic, with the cat enjoying a quiet moment of rest under the covers.

## Brief Description

Q: `<image>`\n Describe the image concisely.

A: A cat is sitting on a wooden bench.

## Complex Reasoning

Q: `<image>`\n Why is the cat positioned in a specific way on the bench?

A: The cat is positioned with its head up and eyes open, lying comfortably on the bench.

Sample multimodal interaction with fMRI data. Image is for reference only.

# Natural Language Interaction- Comparison with SOTA

Table 1: Quantitative analysis of brain captioning, detailed descriptions, and complex reasoning tasks. Some results are derived from UMBRAE [24]. Results are compared to those from other studies, with **best**, **second**, and **third** highlighted. Underline indicates the best result under identical conditions, while \* denotes results obtained using BLIP2-generated captions as ground truth.

| Method             | # Models | BLEU1 | BLEU2 | BLEU3 | BLEU4 | METEOR | ROUGE | CIDEr  | SPICE | CLIP-S |
|--------------------|----------|-------|-------|-------|-------|--------|-------|--------|-------|--------|
| Brain Caption      |          |       |       |       |       |        |       |        |       |        |
| SDRecon [9]        | 4        | 36.21 | 17.11 | 7.72  | 3.43  | 10.03  | 25.13 | 13.83  | 5.02  | 61.07  |
| OneLLM [38]        | 4        | 47.04 | 26.97 | 15.49 | 9.51  | 13.55  | 35.05 | 22.99  | 6.26  | 54.80  |
| UniBrain [39]      | 4        | —     | —     | —     | —     | 16.90  | 22.20 | —      | —     | —      |
| BrainCap [40]      | 4        | 55.96 | 36.21 | 22.70 | 14.51 | 16.68  | 40.69 | 41.30  | 9.06  | 64.31  |
| UMBRAE [24]        | 1        | 57.84 | 38.43 | 25.41 | 17.17 | 18.70  | 42.14 | 53.87  | 12.27 | 66.10  |
| Our Method         | 1        | 57.19 | 37.17 | 23.78 | 15.85 | 18.60  | 36.67 | 49.51  | 12.39 | 65.49  |
| w/o ViT3D          | 1        | 52.91 | 32.18 | 15.64 | 8.49  | 14.07  | 23.25 | 39.64  | 8.34  | 56.92  |
| Our Method*        | 1        | 64.26 | 51.44 | 47.70 | 32.17 | 20.41  | 52.61 | 83.94  | 18.27 | 68.72  |
| w/o ViT3D*         | 1        | 58.87 | 42.11 | 29.48 | 21.39 | 15.85  | 38.48 | 56.37  | 11.27 | 64.35  |
| Detail Description |          |       |       |       |       |        |       |        |       |        |
| Our Method         | 1        | 38.91 | 24.02 | 15.24 | 12.41 | 18.44  | 27.83 | 42.58  | 18.41 | 56.16  |
| w/o ViT3D          | 1        | 33.57 | 18.95 | 11.09 | 6.13  | 15.56  | 23.80 | 20.23  | 16.21 | 51.47  |
| Complex Reasoning  |          |       |       |       |       |        |       |        |       |        |
| Our Method         | 1        | 65.41 | 59.61 | 50.68 | 36.46 | 34.46  | 62.60 | 217.83 | 60.29 | 80.96  |
| w/o ViT3D          | 1        | 60.36 | 47.81 | 39.76 | 30.57 | 24.37  | 45.39 | 150.67 | 52.13 | 73.26  |

# Visual Reconstruction

$$\hat{x} = \mathcal{D}((1 - \beta)\hat{z}_v + \beta\sigma \mid \hat{z}_c, a_r), \quad \sigma \in \mathcal{N}(0, 1).$$

- LLM generates a reconstruction prompt that is combined with the latent representations obtained from the fMRI feature extractor
- These components are given to UnCLIP-2, a generative stable diffusion model



# Visual Reconstruction - impact of B

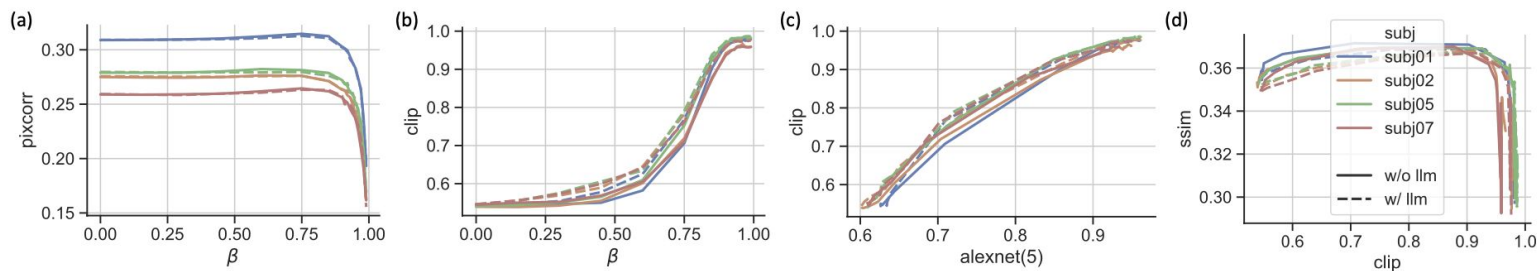


Figure 5: Ablation analysis of the hyperparameter  $\beta$  on visual reconstruction performance.

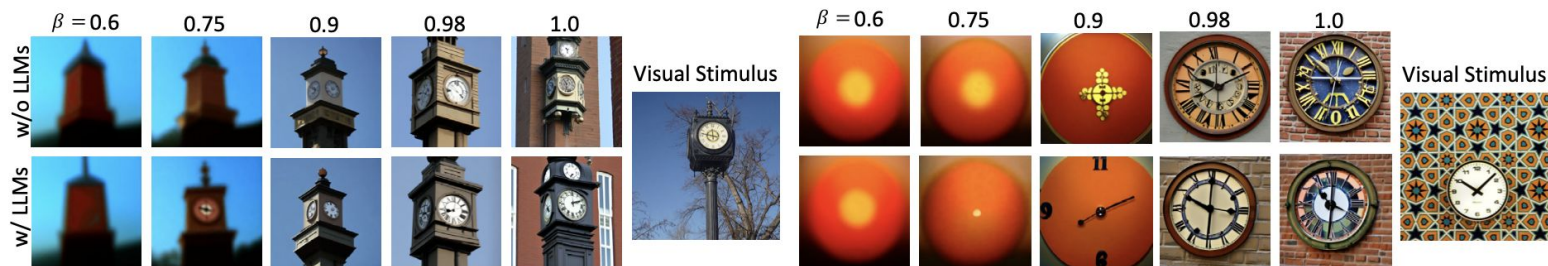
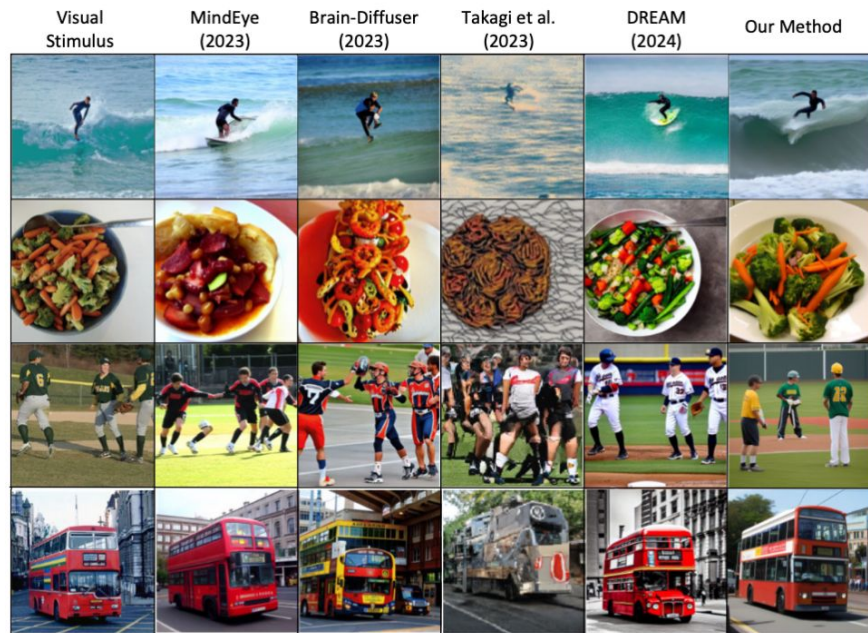


Figure 6: Visualization of the impact of  $\beta$  on visual reconstruction.

# Visual Reconstruction- Comparison Against SOTA



(a) All Trials



(b) Single Trial

# Visual Reconstruction- Comparison Against SOTA

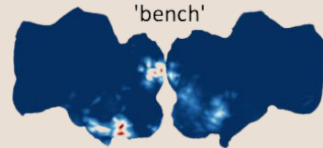
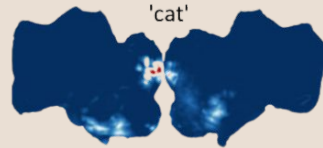
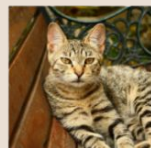
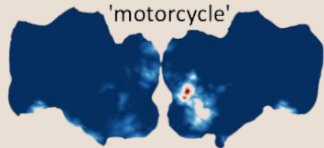
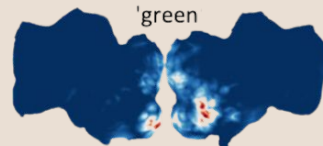
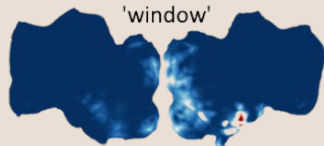
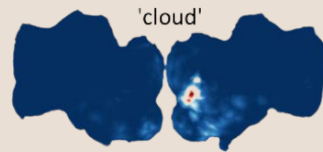
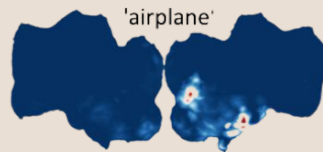
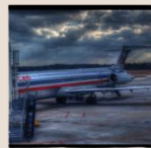
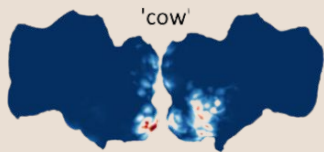
Table 2: Quantitative evaluation on visual reconstruction. Performance metrics are reported across different levels of features, with the **best**, **second** and **third** scores highlighted. The underline indicates the best result under the same conditions. Our method achieves state-of-the-art results using a single model trained across subjects (# Models = 1).

| Method                  | # Models | Low-Level |        |              |              | High-Level  |        |            |        |
|-------------------------|----------|-----------|--------|--------------|--------------|-------------|--------|------------|--------|
|                         |          | PixCorr ↑ | SSIM ↑ | AlexNet(2) ↑ | AlexNet(5) ↑ | Inception ↑ | CLIP ↑ | EffNet-B ↓ | SwAV ↓ |
| Mind-Reader [21]        | 4        | —         | —      | —            | —            | 78.2%       | —      | —          | —      |
| Takagi <i>et al</i> [9] | 4        | —         | —      | 83.0%        | 83.0%        | 76.0%       | 77.0%  | —          | —      |
| Gu <i>et al</i> [41]    | 4        | .150      | .325   | —            | —            | —           | —      | .862       | .465   |
| Brain-Diffuser [42]     | 4        | .254      | .356   | 94.2%        | 96.2%        | 87.2%       | 91.5%  | .775       | .423   |
| MindEye [6]             | 4        | .309      | .323   | 94.7%        | 97.8%        | 93.8%       | 94.1%  | .645       | .367   |
| DREAM [11]              | 4        | .288      | .338   | 95.0%        | 97.5%        | 94.8%       | 95.2%  | .638       | .413   |
| MindBridge [25]         | 1        | .151      | .263   | 87.7%        | 95.5%        | 92.4%       | 94.7%  | .712       | .418   |
| UMBRAE [24]             | 1        | .283      | .328   | 93.9%        | 96.7%        | 93.4%       | 94.1%  | .700       | .393   |
| Our Method              | 1        | .265      | .357   | 93.1%        | 97.1%        | 96.8%       | 97.5%  | .633       | .321   |
| w/o LLM                 | 1        | .263      | .369   | 92.0%        | 97.1%        | 94.2%       | 96.1%  | .680       | .328   |
| w/o VAE feature         | 1        | .093      | .263   | 84.5%        | 90.6%        | 93.6%       | 95.7%  | .684       | .398   |
| w/o C_Subj              | 4        | .241      | .356   | 88.1%        | 95.7%        | 92.1%       | 95.1%  | .631       | .347   |
| w/o C_Subj & ViT3D      | 4        | .164      | .273   | 86.7%        | 91.4%        | 89.3%       | 91.8%  | .731       | .417   |
| Single Trial            |          |           |        |              |              |             |        |            |        |
| MindEye [6]             | 4        | .255      | .308   | 91.6%        | 95.9%        | 91.3%       | 91.6%  | .691       | .398   |
| Our Method              | 1        | .257      | .336   | 91.2%        | 96.3%        | 94.6%       | 95.3%  | .671       | .324   |

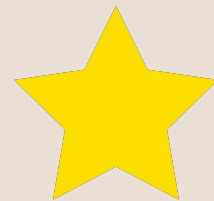
# Concept Localization

- Use GradCAM to generate heatmaps that represent regions of the brain activated in response to the visual stimuli

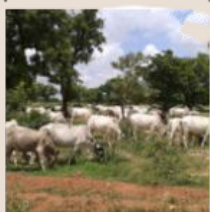
Visual Stimulus



# Concept Localization



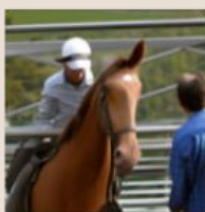
Visual Stimulus



Reconstruction



original



- 'cow'



- 'tree'

Visual Stimulus



Reconstruction



original



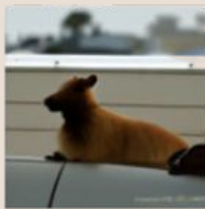
- 'kitchen'



- 'window'



original



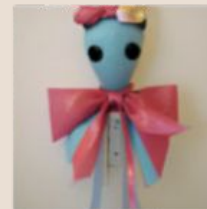
- 'cat'



- 'bench'



original



- 'bear'



- 'toy'

Figure 8: Validation of concept localization by semantic signal nullification and its effect on visual reconstruction.

# Conclusion

- The authors design a framework for fMRI visual reconstruction and language interaction
- Limitations include:
  - generalization across diverse population
  - computational complexity and resource requirements
  - challenges in real world applications

**Thank you! Questions?**